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https://gabriel-vzh2vs.github.io/SYS3062-Website/

UVA's Systems and Information Engineering

March 23rd, 2026

SYS 3062: Lab 7

Compartmental Models Lab

Background: Compartmental Models

- **Definition:** A system dynamics modeling technique that analyzes the movement of entities (populations, mass, etc.) through discrete, homogeneous states.
- **Core Mechanism:** Relies on the principle of mass balance, tracking the rate of flow into and out of compartments.
- **Broad Applications:** Operations research (queue networks), epidemiology (disease states), and Pharmacokinetics (drug distribution).

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System Equilibrium: Steady-State Analysis

Concept: A system reaches steady-state when the state variables (e.g., drug mass, infected population) no longer change with respect to time.

Mathematical Condition:

$$\frac{d\mathbf{x}}{dt} = f(\mathbf{x}) = 0$$

where $\mathbf{x}$ is the vector of state variables.

Application in PK (Continuous Infusion): If a drug is administered via continuous IV infusion at rate R_0, the one-compartment model becomes:

$$\frac{dX}{dt} = R_0 - k_{el}X$$

At steady state ($\frac{dX}{dt} = 0$), the steady-state amount is $X_{ss} = \frac{R_0}{k_{el}}$.

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System Stability: The Jacobian Matrix

How do we know if our simulation will converge to a steady state or diverge into numerical instability? We analyze the **Jacobian Matrix (J)** of the system of ODEs.

For a two-dimensional system $\mathbf{x} = [x_1, x_2]^T$:

$$J = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix}$$

Stability Criterion: Evaluate the eigenvalues (λ) of J at the steady-state point.

- If the real parts of all eigenvalues are negative ($Re(\lambda_i) < 0$), the system is asymptotically stable (perturbations decay).
- Linear PK compartmental models are inherently stable, allowing for reliable long-term simulation.

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Determining Steady-State in Simulation

In practice, how do we programmatically detect steady-state during a simulation run without analytical solving?

- **Burn-in Period:** Run the simulation for a sufficiently long time ($t \to \infty$) until transients decay. In PK, 5 half-lives ($\approx 5 \times \frac{0.693}{k_{el}}$) reaches 97% of steady state.
- **Tolerance Checking (Algorithmic):** Implement a rolling check where the derivative approaches zero within a specified tolerance ϵ:

$$|x(t) - x(t - \Delta t)| < \epsilon$$

- **Why it matters:** Prevents unnecessary computational overhead by terminating the integrator once equilibrium is reached.

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The SIR Model (Epidemiology)

A foundational example of compartmental modeling is the SIR model (Kermack-McKendrick, 1927), which tracks population states using non-linear ODEs:

$$\frac{dS}{dt} = -\frac{\beta}{N} IS$$
$$\frac{dI}{dt} = \frac{\beta}{N} IS - \gamma I$$
$$\frac{dR}{dt} = \gamma I$$

Parameters:

- N: Total population size ($N = S + I + R$)
- β: Transmission rate (probability of disease transmission)
- γ: Recovery rate (inverse of average infection duration)

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Pharmacokinetics (PK) Modeling

- **Goal:** Model the time-course of drug absorption, distribution, metabolism, and excretion (ADME) within a biological system.
- **The Analogy:** Just as the SIR model tracks individuals moving through disease states, PK models track drug mass moving through physiological volumes.
- **Method:** Uses a deterministic framework defined by coupled, linear Ordinary Differential Equations (ODEs).

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Task 1: The One-Compartment Model

Concept: The body acts as a single, kinetically homogeneous unit (e.g., IV Bolus). The drug mixes instantaneously.

Governing Differential Equation:
By mass balance of the drug amount (X), and assuming a constant Volume of Distribution (V_d), where $X = V_d \cdot C_p$:

$$\frac{dX}{dt} = -k_{el}X \implies \frac{dC_p}{dt} = -k_{el}C_p$$

where k_{el} is the elimination rate constant and C_p is plasma concentration.

Analytical Solution:

$$C_p(t) = C_0 e^{-k_{el}t} \implies \ln C_p = \ln C_0 - k_{el}t$$

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Task 1: Simulation Requirements

1. **Develop Model:** Implement $\frac{dC_p}{dt} = -k_{el}C_p$ in your simulation environment.
2. **Simulate:** Run the numerical integration for 10 hours using:
   - Initial Concentration (C_0): $100\ \mu$g/mL
   - Elimination Rate (k_{el}): 0.347 h^{-1}
3. **Verify:** Plot your numerical results against the analytical solution ($C_p(t) = 100e^{-0.347t}$) to verify your integrator's accuracy.
4. **Extract Parameters:** Plot $\ln(C_p)$ vs. time and verify the slope corresponds to $-k_{el}$.

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Algorithm 1: One-Compartment Model (Part 1: Integration)

Objective: Set up the initial value problem and numerically integrate the one-compartment model.

Require: Initial concentration C_0, Elimination rate k_{el}, End time T
Ensure: Time vector t, Simulated concentration C_{sim}

▷ **Step 1: Define System and Integrate**

1: Define ODE: $f(t, C) \leftarrow -k_{el} \cdot C$
2: $(t, C_{sim}) \leftarrow \text{ODESolver}(f, C_0, [0, T])$ ▷ e.g., RK45

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Algorithm 1: One-Compartment Model (Part 2: Verification)

Objective: Generate the analytical solution and verify the simulated parameters via log-linear regression.

Require: t, C_{sim}, C_0, k_{el}
Ensure: Analytical C_{ana}, Estimated $\hat{k}_{el}$ and $\hat{C}_0$

▷ **Step 2: Analytical Solution**

for each $t_i \in t$ **do**
    $C_{ana}[i] \leftarrow C_0 \cdot \exp(-k_{el} \cdot t_i)$
end for

▷ **Step 3: Parameter Extraction via Regression**

$\ln C_{sim} \leftarrow \ln(C_{sim})$ ▷ Log-transform simulated data
$(\text{slope}, \text{intercept}) \leftarrow \textsc{LinearRegression}(t, \ln C_{sim})$
$\hat{k}_{el} \leftarrow -\text{slope}$
$\hat{C}_0 \leftarrow \exp(\text{intercept})$

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Task 2: Two-Compartment Model

Concept: Acknowledges body heterogeneity and delayed drug distribution.

- **Central Compartment (1):** Blood and highly perfused organs. Elimination occurs solely from here. Volume $= V_1$.
- **Peripheral Compartment (2):** Poorly perfused tissues. Volume $= V_2$.

Coupled ODEs (Based on Amount, X):
Note: We must use amounts because $V_1 \neq V_2$.

$$\frac{dX_1}{dt} = k_{21}X_2 - (k_{12} + k_{el})X_1$$
$$\frac{dX_2}{dt} = k_{12}X_1 - k_{21}X_2$$

Concentration is then found via $C_1(t) = X_1(t)/V_1$.

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Task 2: Simulation Requirements

Analytical Solution (Central Concentration): $C_1(t) = Ae^{-\alpha t} + Be^{-\beta t}$

1. **Implementation:** Define the system of coupled ODEs in your simulation software.
2. **Phase Analysis:** Simulate the system and generate a semi-log plot ($\log C_1$ vs t).
3. **Observation:** Identify the bi-phasic curve:
   - Initial rapid decline (Distribution Phase / α-phase)
   - Terminal slower decline (Elimination Phase / β-phase)
4. **Method of Residuals (Curve Stripping):** Decompose the simulated curve into its two linear components to estimate the intercept constants A and B.

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Algorithm 2: Bi-Phasic Simulation (Part 1: Integration)

Objective: Simulate mass transfer in a two-compartment model respecting physiological volumes.

Require: $k_{12}, k_{21}, k_{el}, V_1, \text{Dose}, T$
Ensure: Phase constants (α, β) and intercept constants (A, B)

```
function dX_dt(t, X_1, X_2)                    ▷ Define Mass Balance ODEs
    rate_1 ← k_21 · X_2 − (k_12 + k_el) · X_1
    rate_2 ← k_12 · X_1 − k_21 · X_2
    return [rate_1, rate_2]
end function
```

▷ **Step 1: Numerical Integration**

$(t, X) \leftarrow \text{ODESolver}(\text{d}X_\text{d}t, [\text{Dose}, 0], [0, T])$

$C_1 \leftarrow X_1/V_1$ ▷ Convert Central Mass to Concentration

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Algorithm 2: Method of Residuals (Part 2: Terminal Phase)

Objective: Isolate the late-stage elimination phase to find β and B.

Require: Simulated time t, Central Concentration C_1

▷ **Step 2: Isolate Terminal Phase (β)**

$t_{term}, C_{term} \leftarrow \textsc{Filter}(t, C_1 \text{ where } t > t_{threshold})$

$\ln C_{term} \leftarrow \ln(C_{term})$

▷ **Step 3: Log-Linear Regression**

$(\text{slope}_\beta, \text{int}_B) \leftarrow \textsc{LinearRegression}(t_{term}, \ln C_{term})$

$\beta \leftarrow -\text{slope}_\beta$

$B \leftarrow \exp(\text{int}_B)$

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Algorithm 2: Method of Residuals (Part 3: Distribution Phase)

Objective: Calculate residuals to isolate the early distribution phase to find α and A.

Require: t, C_1, β, B

▷ **Step 4: Extrapolate and Calculate Residuals**

$t_{early}, C_{early} \leftarrow \textsc{Filter}(t, C_1 \text{ where } t < t_{dist})$

$C_{ext} \leftarrow B \cdot \exp(-\beta \cdot t_{early})$ ▷ Extrapolate terminal line

$C_{res} \leftarrow C_{early} - C_{ext}$ ▷ Subtract to find residuals

▷ **Step 5: Fit the Distribution Phase (α)**

$\ln C_{res} \leftarrow \ln(C_{res})$

$(\text{slope}_\alpha, \text{int}_A) \leftarrow \textsc{LinearRegression}(t_{early}, \ln C_{res})$

$\alpha \leftarrow -\text{slope}_\alpha$

$A \leftarrow \exp(\text{int}_A)$

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Your Turn

Now, it's your turn to implement ODE modeling in Python!

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Feedback Form

Throughout the course, I may ask for your feedback on:

- **Course Materials**
- **Lecture & Lab**
- **Assignments**

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Action Items

First, I will thank you all again for coming to lab at 5 pm on a Tuesday!

And now, you need to do the following tasks:

- **Submit** your Python file to Gradescope by next week's lab!
- **Read** Prelab 8 for next week!

Next week we are doing something new!

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